Chapter 8

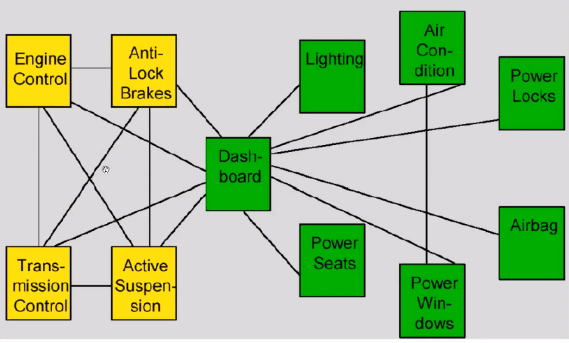
Topic	Control Area Network (CAN) Bus

What is CAN?

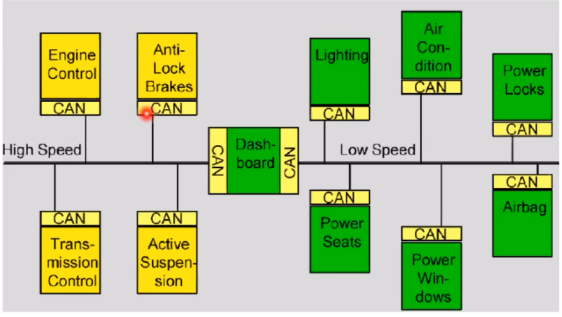
1. CAN connects sensors, actuators, and other control devices to a bus through a host processor and a CAN controller
2. CAN simplifies Physical, Protocol(data link) and Message filtering layer
3. Used for large scale embedded devices

Why do we need CAN?

1. Efficient Communication for vehicles, aerospace, maritime, industrial automation, medical etc use cases
2. Cheap
3. Reliable
4. Simple
5. Standardization

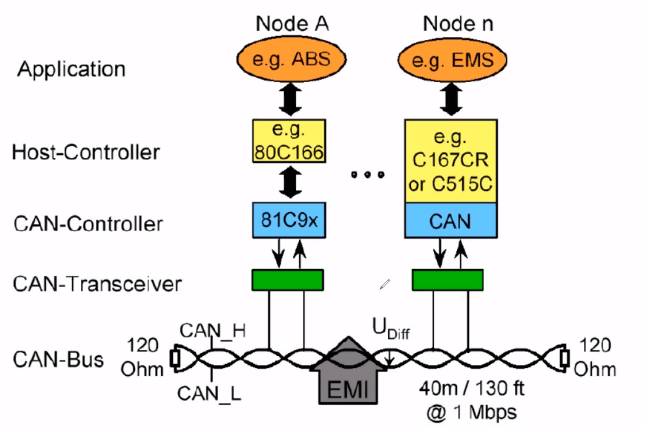


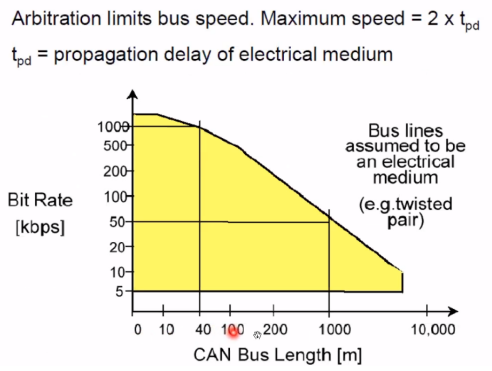
Before CAN, we needed point-to-point interconnections which was not feasible for large networks due to n*(n-1)/2 connections



With CAN, we need only 1 serial bus and CAN specific hardware to each control unit. It is a broadcast type bus (all nodes can hear all transmissions, but message filtering is performed). It uses CSMA/CD to handle collisions

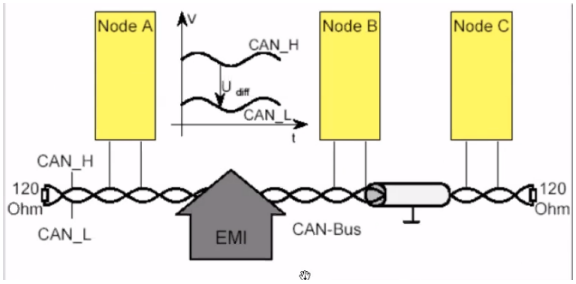
Configuration of CAN



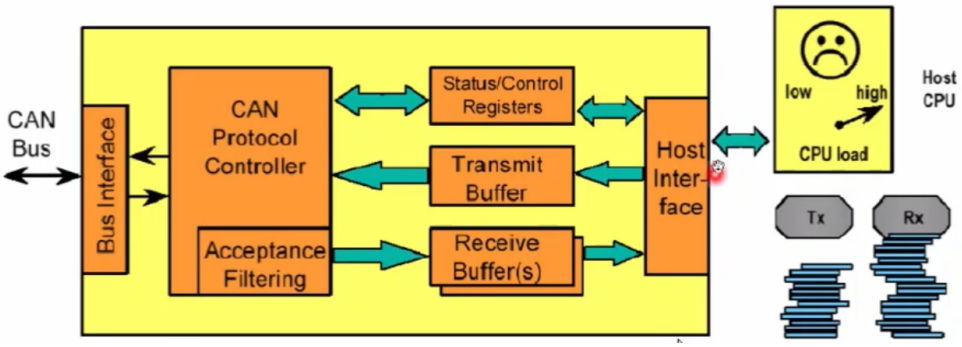


Handling Electromagnetic Interference

To reduce the sensitivity against electromagnetic interference, the bus lines are shielded separated which also reduces electromagnetic emission by itself (useful at high baudrates)



The CAN Controller



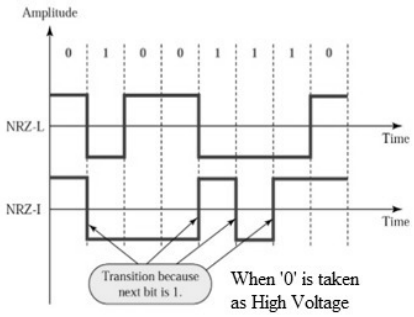
- Host Processor:
  - Decides the meaning of received messages and which messages to transmit to itself
  - Connects with the sensors, actuators and control devices

- CAN controller:
  - A hardware with a synchronous clock
  - Receiving: Stores received bits serially from the bus until the entire message is available. Then the message is fetched by the host processor
  - Sending: Stores the messages to a CAN controller which transmits bits serially onto the bus
- Transceiver:
  - Receiving: adapts signals levels which were received from the bus to the level which the controller expects. It also has protective circuitry
  - Transmitting: converts the transmit bit signal received from the CAN controller into a signal that is sent onto the bus

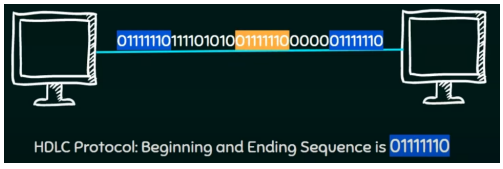
CAN bus features

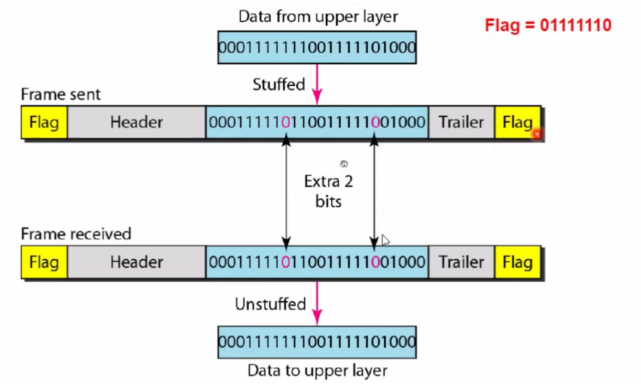
Physical Layer

- Physical layer uses a differential transmission on twisted pair wire
- The bus uses NRZ with bit stuffing
- Nodes are connected using wired-AND fashion (even if only one signal is 0, the output is zero)
- Transfer rate 1000 kbits on 40 meters
- Message length 8 bytes
- Low latency between transmission request and transmission start
- Messages are protected by CRC checksum

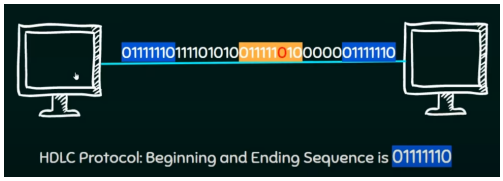


Basic Bit Encoding in CAN bus:





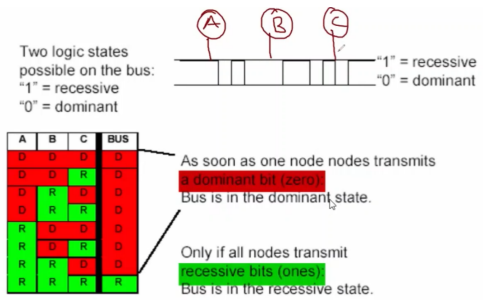
values are wrong, check the next fig for correct values

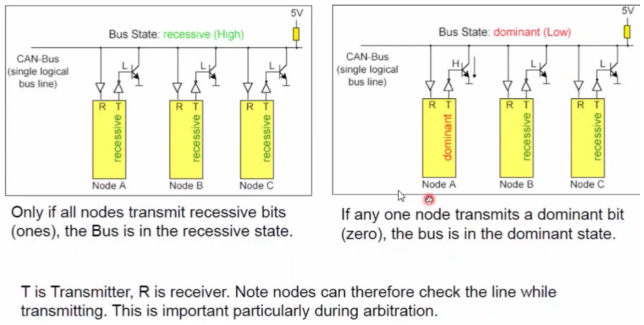


Protocol/Data link Layer

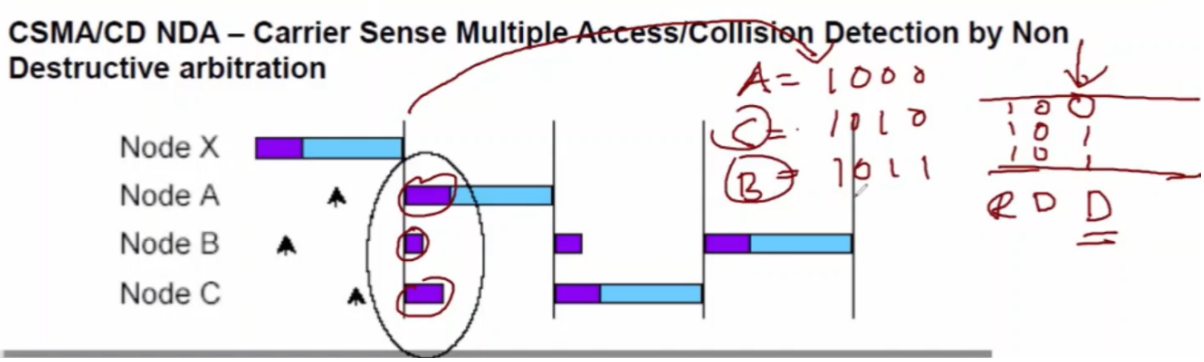
- CSMA/CD with non-destructive arbitration is used
- Each message carries a numeric value which controls the priority on the bus and the identification of the message

CAN bus characteristics(wired-AND fashion)

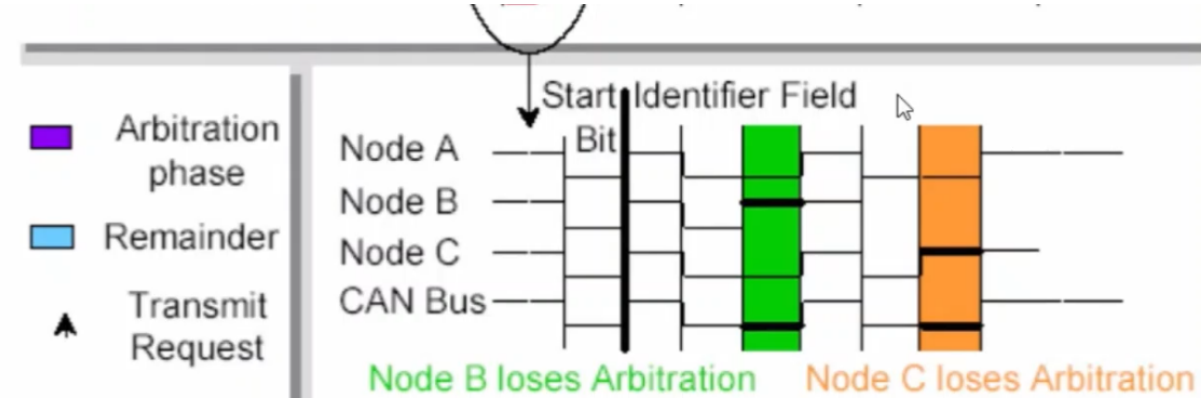




Bus Access and Arbitration



1. All are 1, all are recessive, none has control of the bus
2. All are 0, all are dominant, everyone has control of the bus
3. Even though B and C are 0, the medium is dominant, that means B and C are not dominant node and only A has the control, so A transmits



The CAN protocol

1. Specifies how data packets are transported from point A to point B using a shared medium
2. It does flow control and establishment of communication

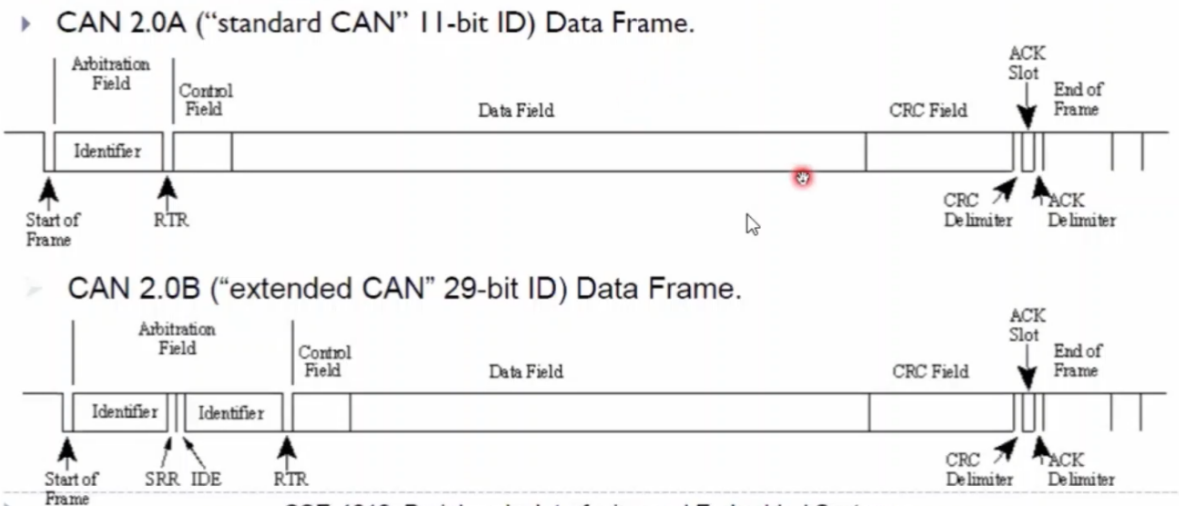
The CAN standard

The Standard defines 4 types of messages

1. Data Frame: most common type of frame
2. Remote Frame: Used for requesting data from a particular node. No data field
3. Error Frame
4. Overload Frame

Shortcut: ***ROED***

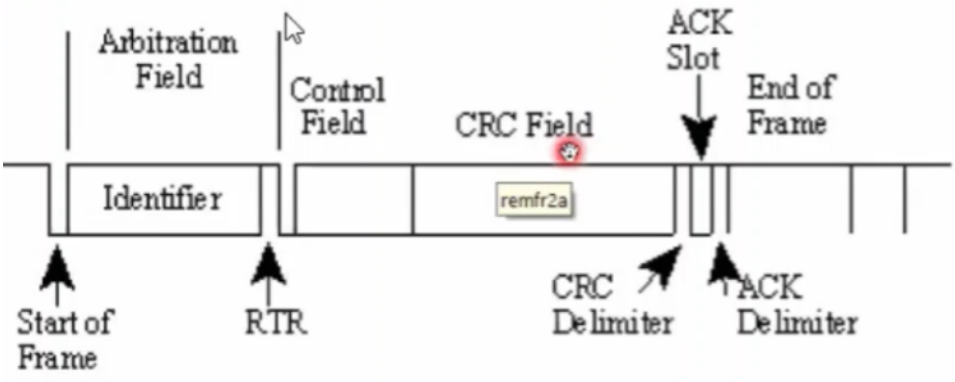
Data Frame



RTR = Request to receive

Remote Frame

2.0A type



Error Frame

Two types of error flags are used

1. Active error flag
2. Passive error flag

These are of 6 dominant bits and is sent by a node detecting active or passive error

Overload Frame

If any node is overloaded with data(receive buffer full) then the node sends this frame to notify it.

Chapter 9

Topic	I^2C (Inter-Integrated Circuit Bus)

What is I^2C (Inter-Integrated Circuit Bus)

1. Used for small scale embedded devices, for a single PCB
2. A serial communication protocol used to connect different components or peripherals in an electronic device
3. uController, EEPROM, LCD driver, A/D converter
4. Speed of 100, 400 and 3400 Kbit/second

Why we need I^2C

1. Can be implemented using fewer pins and smaller packages
2. Simpler architecture
3. Components can be closer to their point of use
4. Cheaper
5. Easy Fault diagnosis
6. No need to design new bus interfaces
7. Multi master slave support

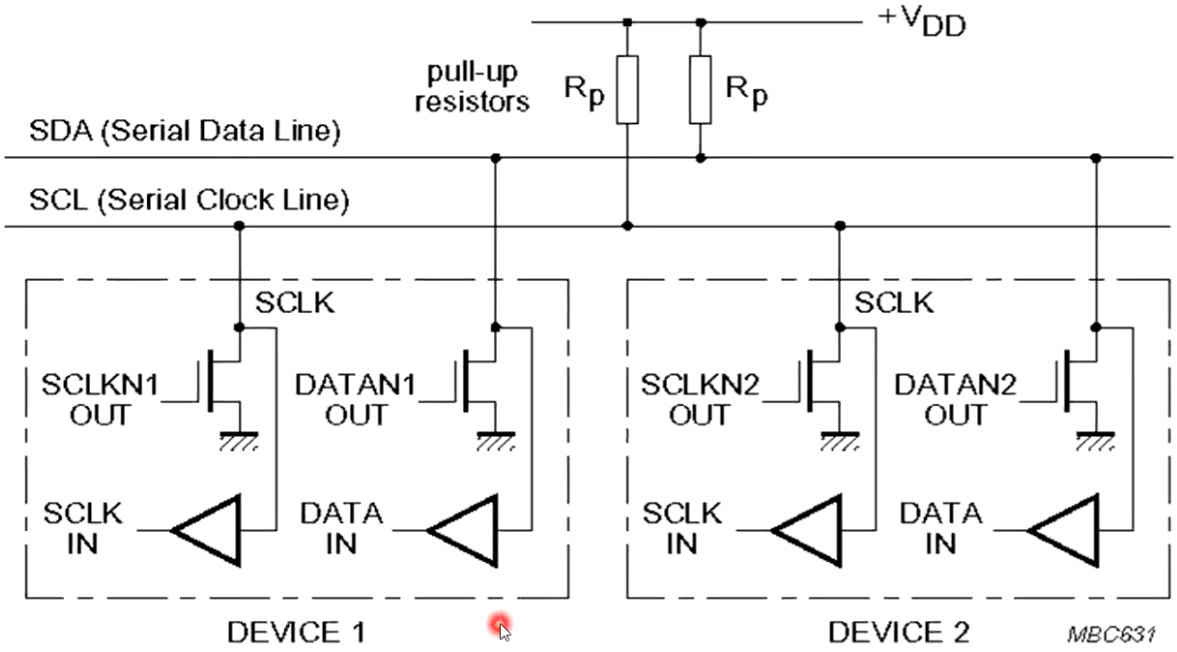
I^2C Bus Characteristics

1. Has features like Synchronous serial signal. It has two wires, one carries data(SDL) while other is used for clock(SCL)
2. Unique start and stop condition
3. Bidirectional data transfer
4. ACK after every transferred byte

5. No fixed length of transfer
6. **Slave selection protocol using 7 bit address**
7. **True Multi Master capabilities**
8. Series resistors for IC protection
9. Compatible with different IC technologies
10. Speed upto 100 kHz

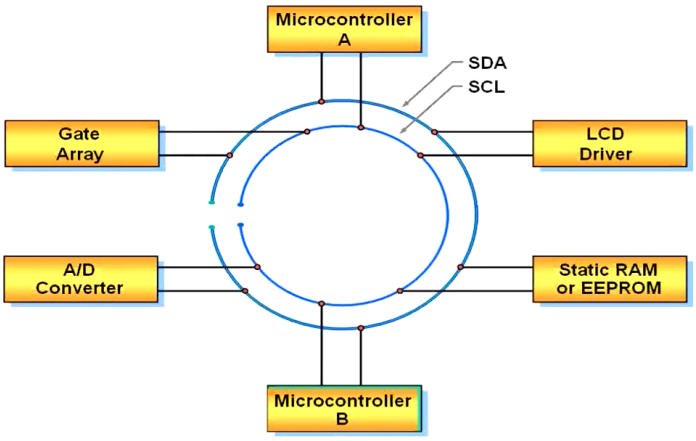


11. Connected devices must have open drain or open collector output for serial clock and data signal
12. All devices must have a common ground reference
13. +5V is supplied
14. I^2C bus can be in idle, master transmit or master receive mode



- **I²C devices are wire ANDed together.**
- **If any single node writes a zero, the entire line is zero**

Master Slave



- **Masters can be**
  - Transmitter only
  - Transmitter and receiver
- **Slaves can be**
  - Receiver only
  - Receiver and transmitter

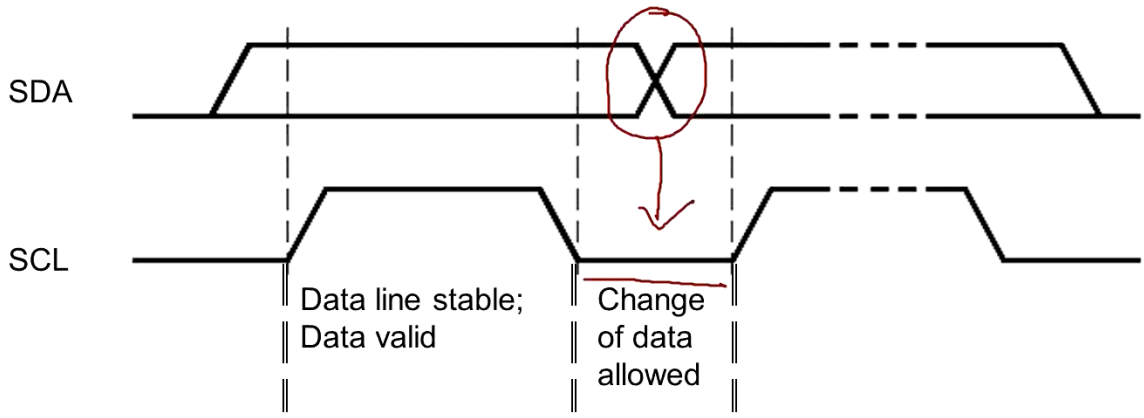
Master

1. Initiates transfer by generating a start and stop condition
2. Generates clock pulses
3. Transmits slave addresses
4. Determines data transfer direction

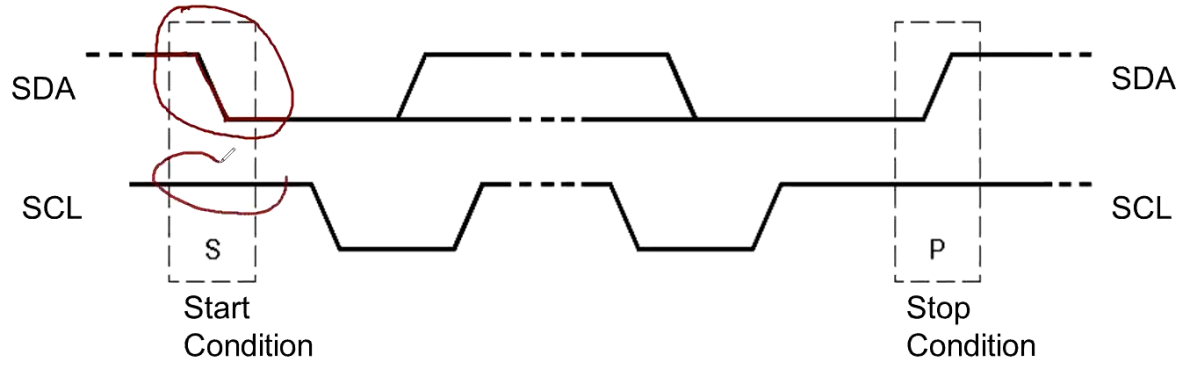
Slave

1. Responds when addressed by master
2. Timing is controlled by clock line

Bit Transfer on the I^2C bus



Normal Data Transfer



Start and Stop condition

When SCL is high but there is a change in SDA

I^2C Addressing

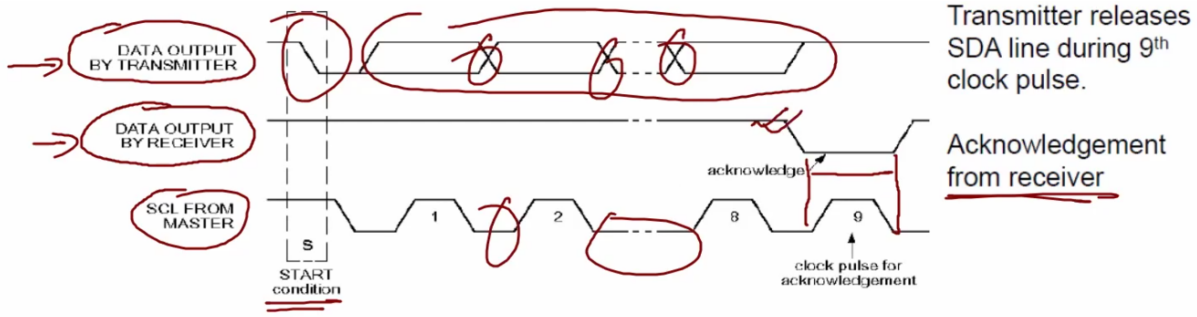
- 0000000 Is a General Call Address
- 0000001 Is a Null (CBUS) Address
- 1111XXX Address Extension
- 1111111 Address Extension – Next Bytes are the Actual Address



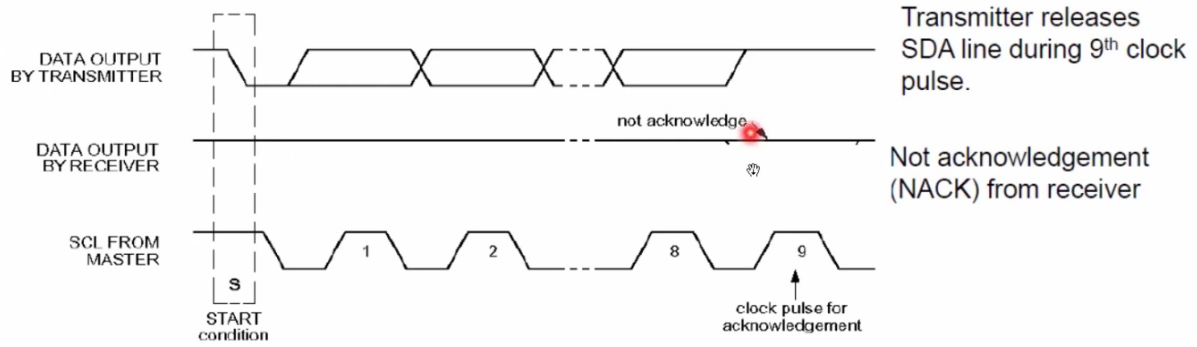
ACK – Generated by the slave whose address has been output.

In data transfer, first 7 bits contain the address of the slave

Handling ACK

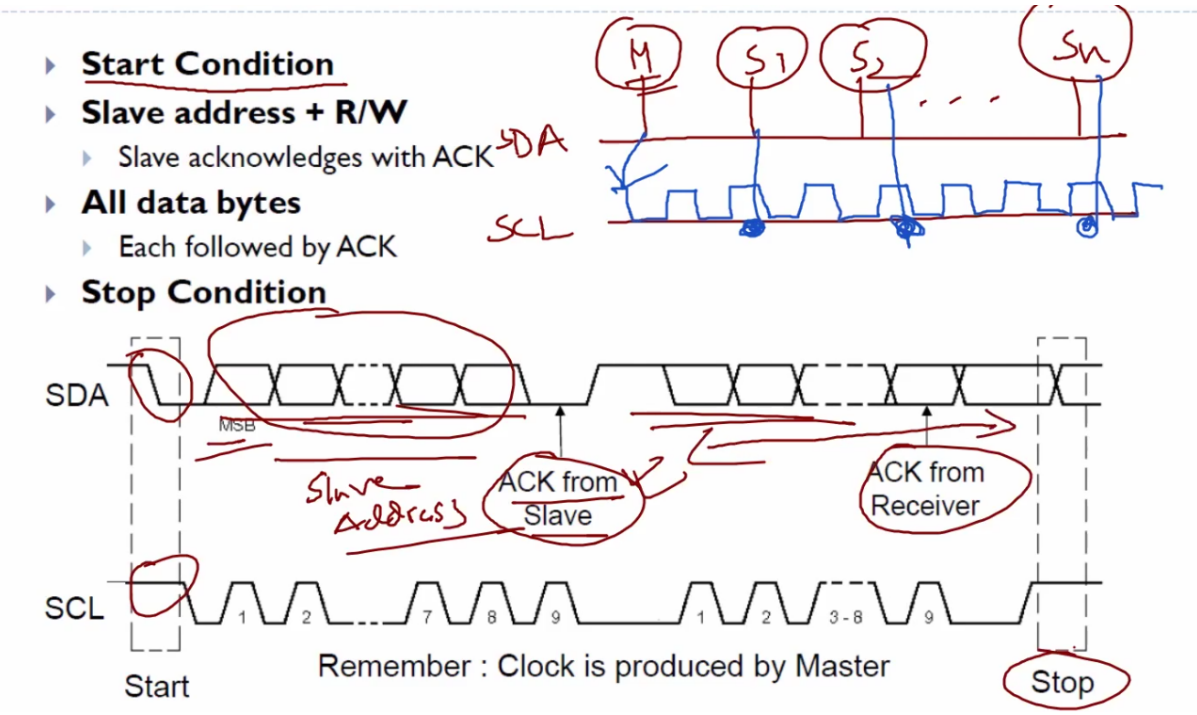


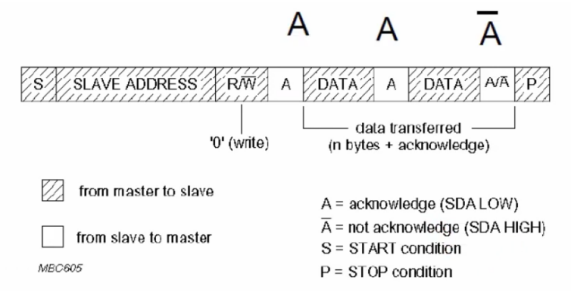
ACK is sent/received upon successful transmission



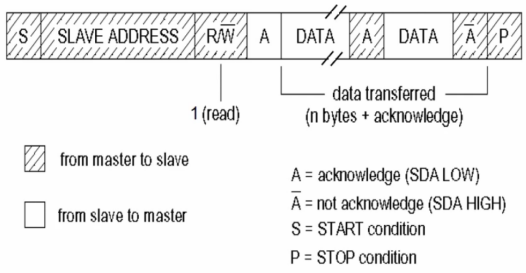
NACK, ACK is sent/received upon unsuccessful transmission

Data Transfer



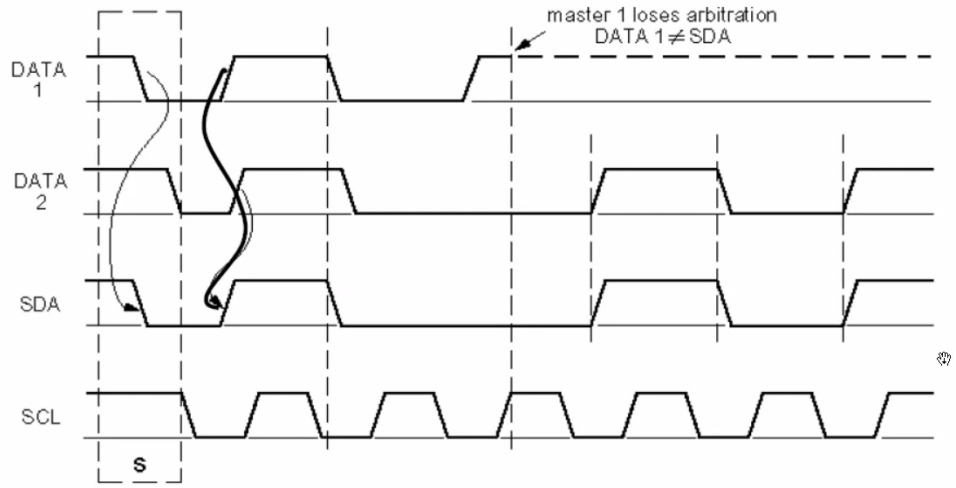


Master writing to slave



Master reading from slave

Arbitration between multiple masters



- As the data line is like a wired-AND, a ZERO address bit overwrites a ONE
- The node detecting that it has been overwritten stops transmitting and waits for the Stop Condition before it retries to arbitrate the bus

The concept is same as before

Error Checking

1. Protocol and timing errors are flagged
2. Glitches can be caused by extra clocks, incorrect data or locked bus
3. A checksum is used to handle these

Bus Recovery

A I^2C bus can be locked when

1. Master and slave are out of sync
2. A stop was omitted
3. Device holding on to the bus line
4. Shorted bus line

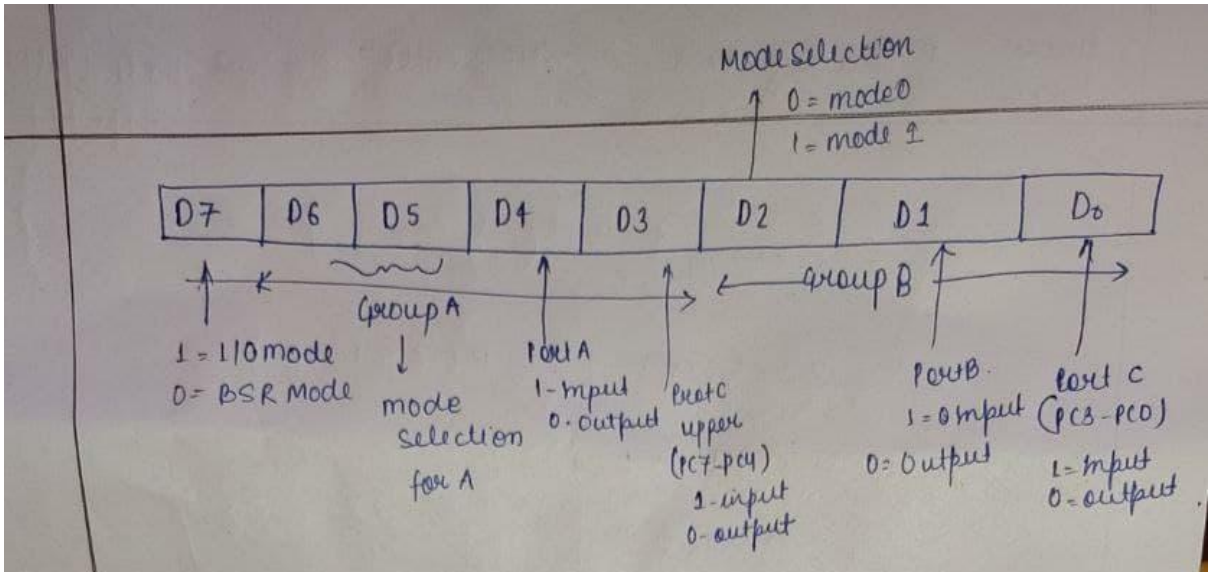
If SCL can be driven, master can send extra clock pulses until SDA becomes high, then a stop.

If SCL is stuck low, only the drive which is driving it can correct the problem

Quiz 2 - 2022

Tags

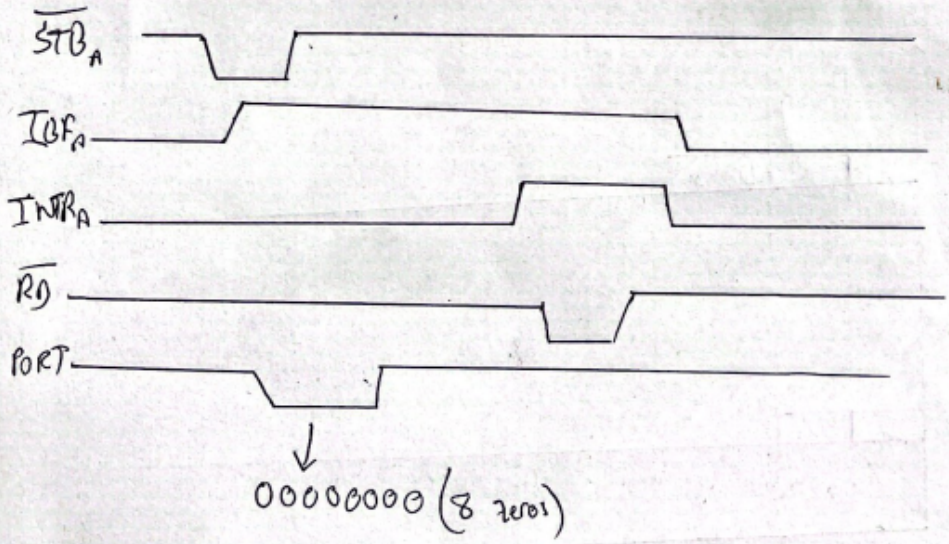
Q1. Draw the timing diagram of a 8255 PPI, when its Port-A is used as input port having interrupt request with a 8-bit data of all '0' and Port-B is used as output port having no interrupt requested with a 8-bit data of all '1'; If Port-C is in the handshaking mode then also derive the command register value. **08**



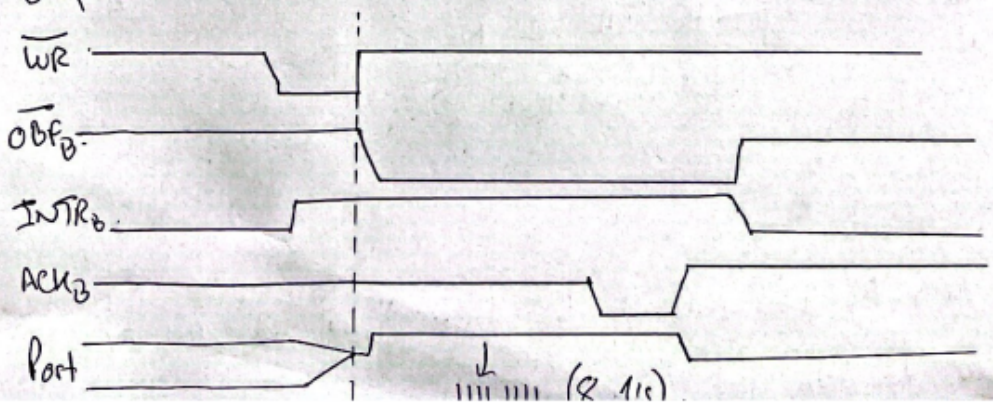
Given the information, the ~~port~~ command register values will be:

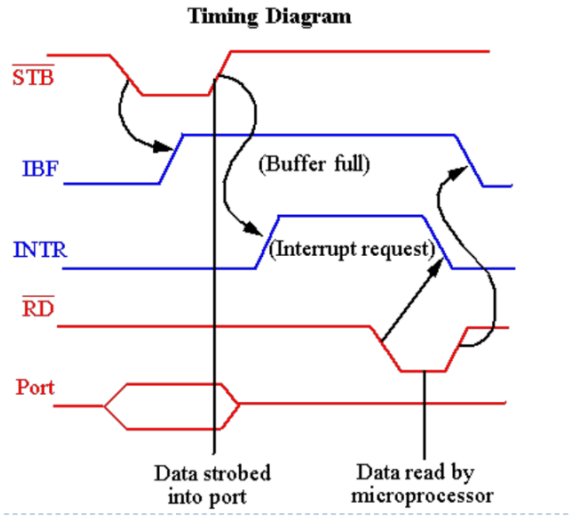
7	6	5	4	3	2	1	0
1	0	1	1	x	1	0	x

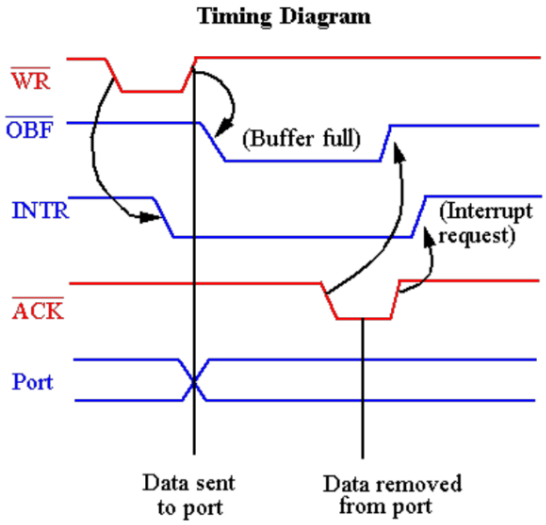
Input:



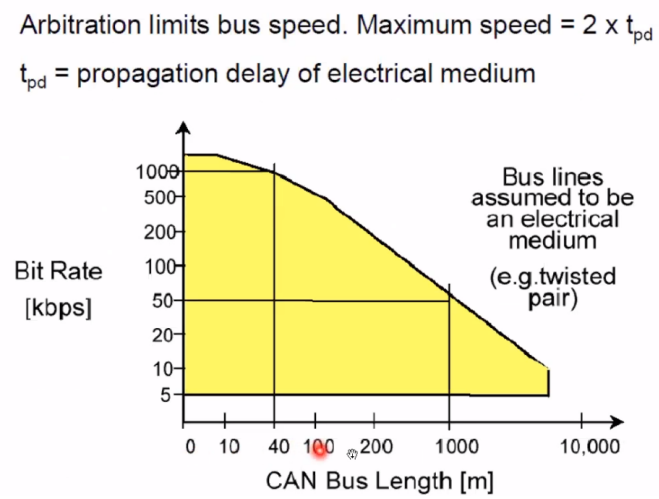
Output:







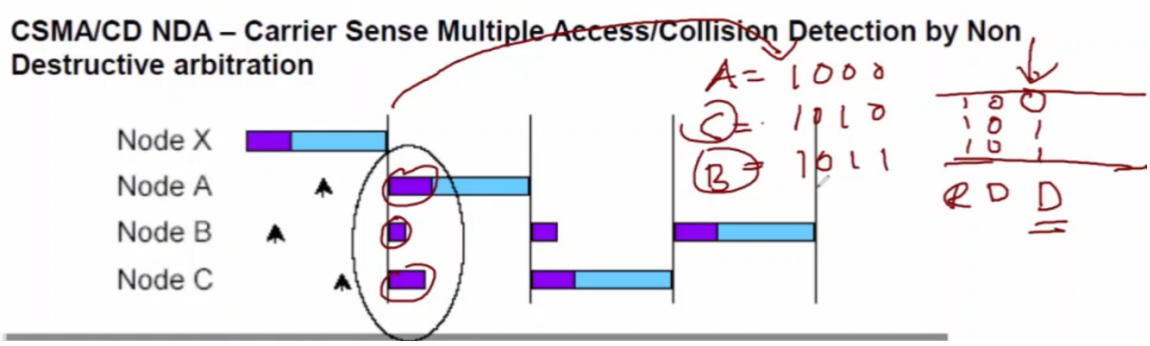
1. What is the maximum length of a CAN bus? How can you justify that the maximum length of CAN bus is appropriate?/ In CAN, with the bus length increment the transmission speed for data rate decreases" – True/False? Justify your answer.



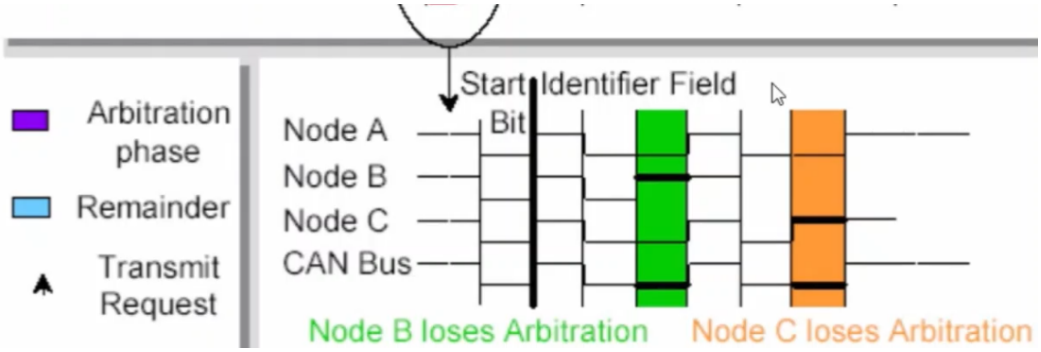
Maximum length is 1km at 40kbps (beyond it, there are drastic decay in quality like 5kbps 5km). But optimum is 40m at 1000kbps. It depends on the application on how long the length of the CAN bus will be. The speed decreases as bus length increases as speed = 2*propagation delay

2. How does the priority of the devices is resolved at CAN bus? Briefly elaborate.

Each message carries a numeric value which controls the priority on the bus and the identification of the message. CSMA/CD with NDA is used

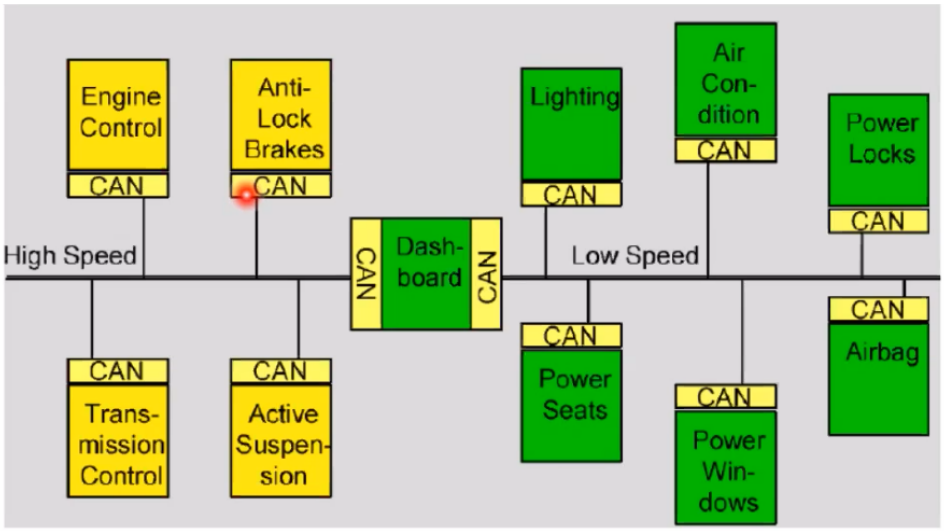


1. All are 1, all are recessive, none has control of the bus
2. All are 0, all are dominant, everyone has control of the bus
3. Even though B and C are 0, the medium is dominant, that means B and C are not dominant node and only A has the control, so A transmits



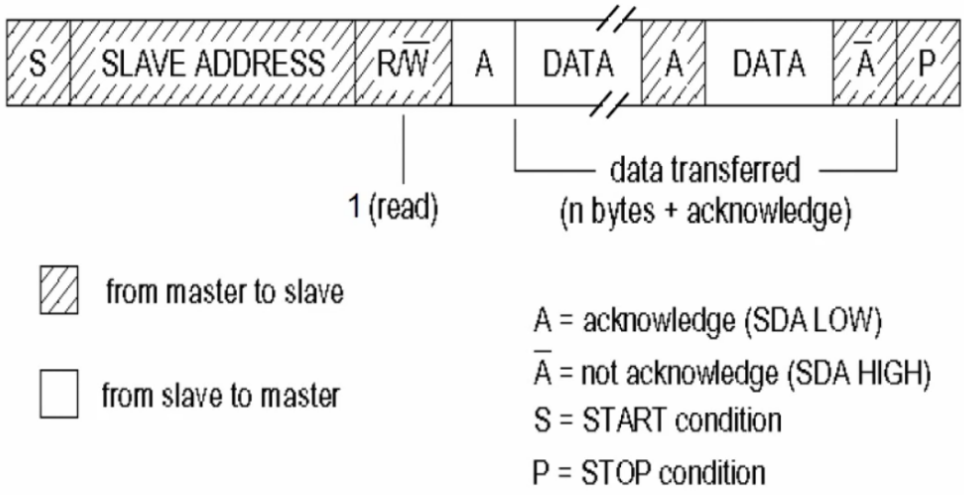
3. Suppose, your student ID is 190041102. Now, for CAN Bus Protocol consider two 07
   component nodes N1 having 11-bit Identifier 101and N2 having 11-bit Identifier 201.
   Using the identifier values draw the timing diagram for CSMA/CD Non-Destructive Arbitration
   (NDA) concept of CAN. Your answer should clearly state the bit-by-bit scenario to show which
   node becomes the dominant to access the CAN Bus and transmit data first.

1. What is CAN bus and why is it called a broadcast type bus? “CAN bus protocol remove $\frac{n(n-1)}{2}$ connections complexity for an embedded system” – explain how?



With CAN, we need only 1 serial bus and CAN specific hardware to each control unit. It is a broadcast type bus (all nodes can hear all transmissions, but message filtering is performed). It uses CSMA/CD to handle collisions

2. Draw the data format of I^2C protocol considering a Slave IC needs to transmit data to the
   Master IC.



Master reading from slave(transmitting)

6. "In I^2C bus connections, master bus can only be transmitter or transmitter-receiver whereas
   slave buses can only be receiver or receiver-transmitter" – Why?

   This is because the slave devices do not initiate communication on the bus and can only respond to the commands sent by the master device. The slave devices

may send data back to the master device as part of their response, in which case they act as receiver-transmitters.

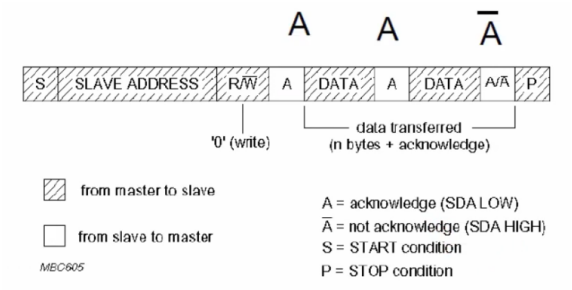
7. Why does in I^2C bus the Start-End condition and Data-Transition signaling are opposite to
each other? Explain.

   The reason why the Start and Stop conditions and the Data transition signaling are opposite to each other is because the data line needs to be released to allow for the Start and Stop conditions to be generated. When the data line is in a stable state (either high or low), the master device can generate the Start condition to signal the beginning of a data transfer. After the Start condition is generated, the data line is released to allow for the slave device to pull the line to either high or low to indicate the bits being transmitted.

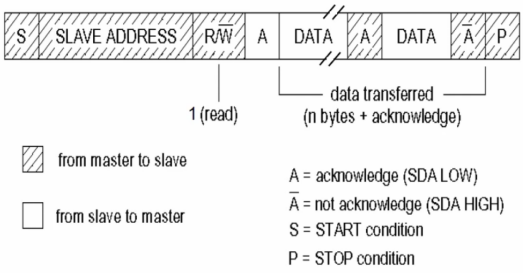
   Similarly, when the data transfer is complete, the master device generates the Stop condition, which indicates that the data transfer is complete. The data line is then released to allow for the line to return to a stable state.

   By using this method of opposite signaling, it ensures that the Start and Stop conditions are clearly defined and separated from the data being transferred. This improves the reliability of the communication on the I2C bus and reduces the chances of errors during data transfer.

7. Draw the data formats of I^2C protocol when the Master IC reads and writes to/from Slave IC



Master writing to slave



Master reading from slave

Merge these frames together